Drawbacks of simple carburettor:

1. Starting difficulty:

To start the engine rich mixture is needed to the engine, but simple carburettor will supply only the lean mixture.

To supply rich mixture we need to use any of the following setup adjustable area jet (or) separate air passage.

2. Idling difficulty:

After the engine starts without the movement of the vehicle the engine alone working is a stage called "idling stage". To maintain the engine at idling stage the suction in the venturi is very less so that it can't take required amount of fuel from the main nozzle. So that there is difficulty to run engine in slow speed.

3. Running difficulty:

When the engine speed is high (or) getting slow, the simple carburettor won't work properly. So to avoid the running difficulty we need to adopt any one of the following methods.

1) Extra air compensation valve
2) Restrict air fuel compensation valve
3) Jet compensation valve
4) Main jet compensation valve.

4. Acceleration difficulty:

When the throttle valve opens suddenly more amount of air is sucked inside, so that there is a delay in petrol supply. So that the lean mixture will be supplied to the engine so the engine struggles at load. To compensate this acceleration pump is being used.

5. Weather difficulty:

If a carburettor is set in hot climate it will give lean mixture during cold climate. In the same way if it is set at cold climate it will give rich mixture during hot climate. To overcome this problem a climatic control device is being installed in advanced carburettors.

6. Icing difficulty:

The venturi part in the carburettor is the place where the petrol gets vaporized. So at that place the cold region is formed by removing hot region. So that at the cold climates and in hill regions the petrol in the carburettor gets freezes. To avoid this idling port and throttle regions of carburettor is heated using exhaust gas, also the hot water from the radiator is poured on to the carburetor which avoids petrol freezing.

7. Altitude difficulty:

At higher altitude regions the pressure will be less which makes air concentration very less. So that will give rich mixture to the engine. Some additional arrangements are made to supply lean mixture to the engine.

9.5.5 Solex Carburettor

To meet the fuel requirments of the different operating conditions of the engine in most of the transportation vehicles like petrol cars the solexcarburator is used. This is one of the down draft type carburetors. With respect to the engines speed and torque the fuel air mixture is prepared by this carburettor and sent through the intake manifold to the

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